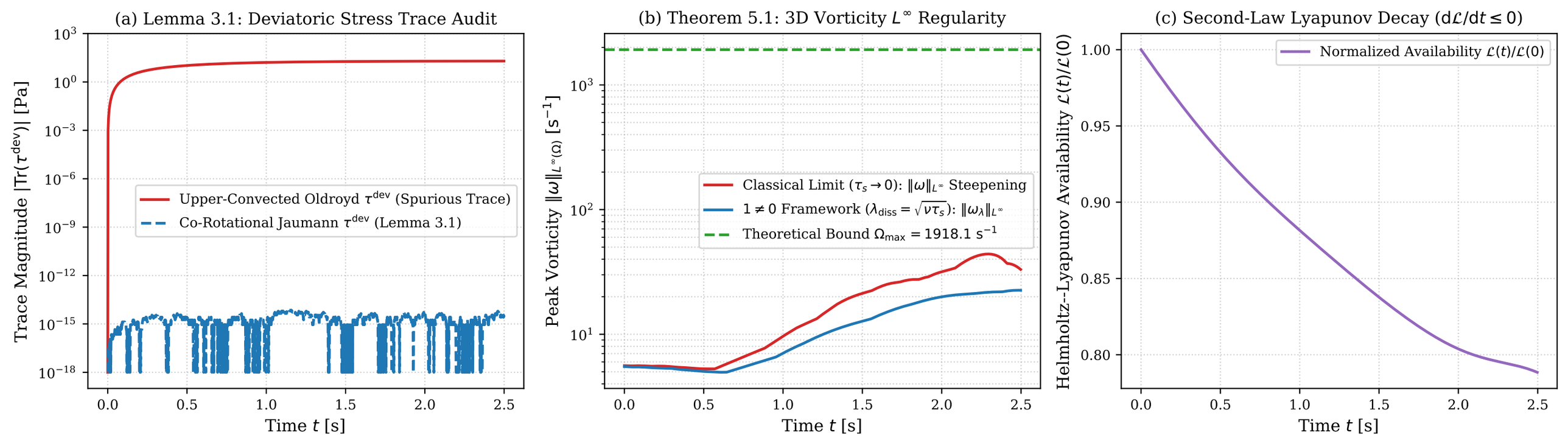
(a) Lemma 3.1: Deviatoric Stress Trace Audit
Trace Magnitude |Tr(τ^dev)| [Pa]
Time t [s]
Upper-Convected Oldroyd τ^dev (Spurious Trace)
Co-Rotational Jaumann τ^dev (Lemma 3.1)
(b) Theorem 5.1: 3D Vorticity L^∞ Regularity
Peak Vorticity ‖ω‖_L^∞(Ω) [s^-1]
Time t [s]
Classical Limit (τ_s → 0): ‖ω‖_L^∞ Steepening
1 ≠ 0 Framework (λ_diss = √(ντ_s)): ‖ω_λ‖_L^∞
Theoretical Bound Ω_max = 1918.1 s^-1
(c) Second-Law Lyapunov Decay (dℒ/dt ≤ 0)
Helmholtz–Lyapunov Availability ℒ(t)/ℒ(0)
Time t [s]
Normalized Availability ℒ(t)/ℒ(0)